Vama Mitesh Soni

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.50
S.M.T LP Savani Vidhyabhawan	Class XII (GSHSEB)	2020	72.46%
J.H. Ambani Saraswati Vidhyamandir	Class X (CBSE)	2018	90.80%

INTERNSHIP EXPERIENCE

Associate Software Machine Learning Engineer - Binoloop Inc., Calgary, Canada

- Designed a **RAG-based LLM system** to automate permitting, procurement and compliance workflow
- **Optimized GCP-based AI workflows** and supported model tuning through statistical analysis.
- Curated RAG evaluation datasets to **benchmark retrieval accuracy and hallucination** using **Cohen's Kappa**

Computer Vision Research Intern - Symbiosis Center of Applied AI (SCAAI), Pune, India

- Curated **AOD-4** dataset with **22K images** & **31K annotations**, spanning **4 classes** across diverse environments
- Demonstrated **OBB's tighter localization** over HBB, improving YOLOv8n mAP50 from 0.925 to 0.980
- Achieved **0.980 mAP@50** and **0.786 mAP@50-95** with **YOLOv8n-OBB** for airborne object detection
- Optimized YOLOv8n-OBB via **hyperparameter tuning** and transfer learning across key params like learning rate, batch size

Python Developer Intern - Jemistry Info LLP., Surat, India

- Developed a **real-time chat application** using Django Channels, Web Sockets and SQLite for async communication
- Integrated OpenCV-based **drowsiness detection** using a **68-point facial landmark** predictor into a Flask application

ADDITIONAL EXPERIENCE

OCR Reviewer (Freelance) - AI4Bharat (A Research Lab at IITM), Chennai, India

- Annotated **4500+** pages across Indian languages to build **high-quality OCR datasets**
- Evaluated **TTS model outputs**, identifying pronunciation inconsistencies to improve accuracy and naturalness.
- Led the review for **Gujarati language** to ensure data quality, consistency, and adherence to labeling standards

SELECTED PERSONAL PROJECTS

Re-acquiring Objects under Long-Term Occlusion in Visual Object Tracking

Working under the guidance of Dr. Sumohana S. Channappayya

- Reconstructed AR²-4FV referring-tracking pipeline using open-source Grounding DINO + CLIP
- Architected a zero-shot **language-grounded re-identification system** with dual independent similarity gates
- Investigating **language-grounded semantic memory** for robust object re-acquisition in dynamic visual environments

Insight-Forge: Agentic Analytics & Decision Intelligence Platform

Python | LangGraph | GPT-OSS 120B | DuckDB | SQL | BM25 | Streamlit

- Built a **LangGraph-based agentic analytics platform** converting natural-language business questions into multi-step SQL, statistical analysis, visualizations, and evidence-backed reports across **100K+ e-commerce orders/reviews**
- Implemented **BM25 schema + value retrieval** and an NLSQL pipeline with **DuckDB execution, SQL validation/repair, and structured tool orchestration**, averaging **~4 queries and 7-8 LLM calls/task**
- Integrated deterministic **Welch's t-test, ², correlation, regression, and descriptive statistics** into the agent loop to statistically validate LLM-generated hypotheses

- Built a **critic + provenance layer** linking findings to executable SQL/results and rejecting unsupported numerical claims; generated end-to-end analyses in **~20-35s**

Curse of Dimensionality in MCMC

- Benchmarked **5 MCMC samplers** (NUTS, HMC, emcee, zeus, slice) across **4 posterior geometries** and **8 dimensionalities**
- Evaluated using ESS/s, KL divergence, and Gelman-Rubin convergence over **140** experiments
- Fitted power-law scaling exponents showing gradient-free samplers **degrade 3-10× faster** with dimension

Semiconductor Wafer Image Restoration

- Build an **end-to-end denoising and 2x super-resolution pipeline** for degraded wafer inspection images
- Fit the sensor degradation model from **3,200 real pairs** and used it to **expand training data ~5x** to **15,880**
- Architected a FiLM-conditioned NAFNet-style encoder-decoder **(9.6M params)** with a **6-term composite loss**
- Achieved **28.65 dB PSNR** and **0.78 SSIM** at **13.2 ms/image** on an RTX 4090

Neural Hangman Solver

- Engineered a **BiLSTM + multi-kernel CNN + attention** model for character-level prediction, incorporating global Hangman-state features and probabilistic aggregation across hidden positions
- Designed a **realistic-state training** pipeline using training-set letter-frequency priors, stochastic state generation and hard-state oversampling
- Trained for **80 epochs** using Adam, cosine learning-rate scheduling and cross-entropy loss, achieving **64.88% on held-out validation set** with an average of **3.763 incorrect guesses;** reported score of **66.4558** on leaderboard

Mini-ML Library & Autograd Engine

- Engineered a **reusable NumPy ML package** with **scikit-learn-style** (fit, predict, transform) APIs
- Implemented Logistic Regression (IRLS/Newton-Raphson), LDA, Naive Bayes, Ridge Regression and K-fold
- Built a **reverse-mode autograd engine** to support computation graphs, automatic differentiation, neural-network modules

Culture-Specific NER

- Generated an **88K+ sentence BIO-annotated dataset** using Mistral API with **10** diverse sentences per cultural concept
- Extended the **DIWALI dataset**, enforcing **verbatim concept mentions** for clean span annotation
- Validated dataset quality across **4 dimensions:** grammar, semantic relevance, fluency, lexical diversity using LanguageTool, Sentence-BERT, and GPT-2 perplexity
- Benchmarked CRF, BERT, and MuRIL for culture-specific NER; MuRIL achieved the **best F1-score of 0.849**

ACHIEVEMENTS

3rd Prize, nVisionX'26 Design Challenge

RouteMind: Heterogeneous Fleet Routing Optimization

- Solved NP-hard **heterogeneous capacitated vehicle-routing** problem of Greater Hyderabad
- Built a **Genetic Algorithm + Simulated Annealing + 2-opt solver** across 1,578 collection points and 8-hour shift time
- Achieved **100% service coverage**: 227 routes, 2,116 km, 108 vehicles across a 4T/8T/16T fleet, **~0.84 kg CO2/t**

PUBLICATIONS

Introducing AOD 4: A dataset for air borne object detection, First Author, Data in Brief, July 2024- DOI: *10.1016/j.dib.2024.110801*

SKILLS

Languages: C, C++, Python, SQL, HTML, CSS

ML/DL: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, ML Flow, Hugging Face, OpenCV, YOLO, CNNs, Transformers

Statistical ML: Bayesian Inference, MCMC, Hypothesis Testing, Regression, Convergence Diagnostics, KL Divergence, Cross-Validation, Regularization

AI Engineering: LLMs, Generative AI, RAG, Agentic AI, LangGraph, Tool Calling, Prompt Engineering, Structured Outputs, NL-to-SQL, BM25, Schema Retrieval, Embeddings, Vector Search, Milvus

Data: pandas, NumPy, SciPy, statsmodels, ETL Pipelines, Synthetic Data Generation, Geospatial Data (OSMnx, GeoJSON)

Data Engineering & Databases: PostgreSQL, DuckDB, dbt, Apache Airflow, ETL/ELT, Polaris, PyArrow, Parquet

Visualization: Matplotlib, Seaborn, Plotly, Leaflet.js, Chart.js, Looker Studio

Tools: Git/Github, GitHub Actions, Docker, Roboflow, GCP, Linux, LaTeX, Wireshark, ROS1 Neotic, pytest, Streamlit

RELEVANT COURSES

CS Fundamentals: Object Oriented Programming with C++, Data Structures & Algorithms, Database Management System, Operating Systems, Computer Networks, Software Engineering, Computer Architecture & Organization, Compiler Design

Mathematics: Linear Algebra, Calculus, Probability & Statistics, Discrete Mathematics, Random Variables & Stochastic Processes, Design & Analysis of Algorithms

AI/ML: Foundations of Machine Learning, Deep Learning, Natural Language Processing, Artificial Intelligence, Image Processing & Computer Vision, Data Science Analysis

ADAS: Autonomous Navigation, Internet of Things, Automotive Communication & Sensing

POSITION OF RESPONSIBILITY

Head Placement Coordinator - Center of Interdisciplinary Programs (CIP), IITH

- Leading placement activities for **80+ students** across **7 interdisciplinary programs**
- Coordinating institute-wide initiatives: India-Nagoya Day and Career Fair, engaging **100+** companies across domains
- Lead for SDE & AI/ML and Smart Mobility & Robotics Pitching Cohort '26,
- Coordinating a team of **150+** placement coordinators, driving outreach to **2000+ companies**

Gate Mentor - Maargdarshan

- Mentored **120+** GATE aspirants on college-branch selection based on GATE scores, academics, goals and preferences
- Delivered **1:1 paid mentorship**, aligning postgraduate options with students' long-term career objectives

Lead - Google Developer Students Club, CHARUSAT University

- **Revamped GDSC operations** by streamlining team, peer-learning, and industry partnerships resulting in **6** collaborations
- Led a **25-member team** to organize **10+** events, driving **191.8%** membership growth to **1000+** members
- Led **GDSC WoW 2023's** first offline edition, coordinating **70-member team**, **750+** attendees, and securing sponsorships

HOBBIES & INTEREST

Inter-IIT Athletics 2026 | Acrylic Painting | Classical Dance: Bharatnatyam

1st runner-up in **1500m and 400m** under Women Category in **Institute Athletics League '26**